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Traffic Sign Recognition System Using CNN

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1. ABSTRACT

Traffic sign recognition is a vital component in the development of autonomous vehicles and advanced driver-assistance systems (ADAS). This project report details the development of a traffic sign recognition system using Convolutional Neural Networks (CNNs), a deep learning approach renowned for its effectiveness in image recognition tasks. The system leverages the German Traffic Sign Recognition Benchmark (GTSRB) dataset, preprocessing techniques, and a specifically designed CNN architecture to accurately classify traffic signs into 43 different categories. The implementation includes data augmentation to enhance model robustness, and the training process employs techniques such as early stopping to prevent overfitting. The system achieves high accuracy and reliability, with a validation accuracy of 98.47%, making it suitable for real-time application in autonomous driving. Future enhancements are proposed to further optimize the model for real-time deployment, improve robustness under varying conditions, and extend the dataset for broader applicability. This project demonstrates the potential of CNNs in enhancing traffic sign recognition, contributing to safer and more efficient autonomous vehicle systems.

2. INTRODUCTION

In recent years, the development of autonomous vehicles and advanced driver-assistance systems (ADAS) has gained significant momentum. A crucial aspect of these systems is their ability to recognize and interpret traffic signs accurately and in real-time. Traffic signs provide essential information for safe and efficient driving, such as speed limits, warnings about road conditions, and guidance on the road layout. Effective traffic sign recognition can prevent accidents, ensure compliance with traffic laws, and enhance overall road safety.

Traditional methods of traffic sign recognition relied heavily on hand-crafted features and conventional machine learning techniques. However, these methods often struggle to handle the variability and complexity of real-world traffic sign images, which can vary widely in terms of lighting conditions, angles, occlusions, and wear and tear.

The advent of deep learning, particularly Convolutional Neural Networks (CNNs), has revolutionized the field of image recognition. CNNs are designed to automatically and adaptively learn spatial hierarchies of features through backpropagation, making them exceptionally well-suited for tasks involving image data. They have been successfully applied to a variety of computer vision tasks, including object detection, facial recognition, and, importantly, traffic sign recognition.

This project focuses on the development and evaluation of a traffic sign recognition system using CNNs. The system is built and trained using the German Traffic Sign Recognition Benchmark (GTSRB) dataset, which is a widely used dataset for this purpose, containing over 34,000 images of traffic signs classified into 43 categories.

The project involves several key stages: data preprocessing, CNN model design, model training, evaluation, and testing. By leveraging CNNs, the system aims to achieve high accuracy and robustness in recognizing traffic signs under various conditions. The goal is to create a reliable and efficient traffic sign recognition system that can be integrated into autonomous vehicles and ADAS to improve safety and navigation.

This report provides a comprehensive overview of the project, detailing the literature survey, software requirement specification, detailed design, implementation, testing,

conclusion, future enhancements, and bibliography. Through this systematic approach, the project aims to contribute to the ongoing efforts in enhancing autonomous driving technologies.

3. Literature Review

[1] Traffic sign detection and recognition (TSDR) plays a major role in this as it detects and recognizes traffic signs, thereby alerting a driver if any traffic signs are approaching. By doing so, besides the safety on road being ensured, the driver will feel more relaxed when navigating tricky or unfamiliar roads. Oftentimes, it is difficult to comprehend signs and other warnings. Using this approach, drivers will not have to struggle with translating signs. This study aims to propose a model for TSDR using deep learning wherein convolutional neural networks are used for recognizing traffic signs using a hybrid dataset that comprises German traffic sign recognition benchmark dataset from Kaggle and a self-created Indian traffic sign dataset. In addition, the proposed model is trained on both datasets individually so that the output from the proposed model can be compared with the output of existing models. Experimental results revealed that the proposed model attained an accuracy of 95.45% for hybrid datasets and for the Indian dataset alone 91.08%, whereas for the German dataset alone, accuracy yielded was 99.85%.

[2] Real-Time Traffic Sign Recognition System (RTTSRS) is used for recognizing the traffic signboards (Take left, take right, speed limit 60 kmph… etc.), it plays a crucial role in the domains of driverless vehicles etc. By using Real-Time Traffic Sign Recognition, Traffic related problems can be reduced. It is categorized into two types- localization and recognition. Localization deals with identifying and locating traffic sign regions within the radius. Real-Time Traffic Sign Recognition is used to identify the traffic sign region within the space (rectangular) provided. This study describes an approach for a traffic sign recognition system, many machine learning algorithms like Support Vector Machine (SVM) and Convolutional Neural Networks (CNN) have been studied for recognizing traffic signs. This study has conducted a critical investigation on various machine learning algorithms which gives high accuracy to predict, recognize real-time traffic signs.

[3] Road accidents are a leading cause of fatalities, often resulting from the misinterpretation of traffic signs. Misinterpretation can occur due to driver negligence or reduced visibility caused by adverseweather conditions such as fog or heavy rain. To address this issue and prevent road accidents, we propose an innovative system that employs R-CNN for real-time traffic sign recognition. Our system operates by capturing a live feed from the vehicle's perspective and processing it in real-time. Using R- CNN, it

identifies traffic sign boards and promptly notifies the driver via voice alerts. This system proves particularly valuable when visibility of traffic signs is significantly compromised due to adverse weather conditions. By recognizing and relaying crucial information, it empowers drivers to make informed decisions, thereby reducing the risk of accidents and enhancing road safety.

[4] Automatic traffic sign detection and recognition is crucial and has the potential to be utilized for driver assistance to reduce collisions in driverless cars. A lot of applications in the automotive sector are built around the computer vision challenge of traffic sign detection. The proposed method uses deep Convolutional Neural Networks (CNN) to detect and recognize traffic sign images in real-time. U sing images of the German Traffic Sign Recognition Benchmark (GTSRB) as training data, a sequential CNN model has been built which will be utilized to recognize and classify the unlabeled traffic signs in this challenge. There are 43 classes in the image dataset. The model is implemented in hardware using Raspberry Pi 4 and a web camera.

3.1 Existing System

Traffic signs carry much information necessary for successful driving - they describe up-to-date traffic situations, define right-of-way, prohibit or permit certain actions or directions, warn about risk factors, etc. Road signs also help drivers with routing the vehicle by identifying the road-sign by using computer vision. The road conditions in the actual scene are very complicated so it was really very hard for the researchers to make the system efficient. The existing system has been detected and categorized using standard computer vision methods, but it was taking a long time.

3.2 Proposed System

In this, a traffic sign detection and identification method on account of the image processing is proposed, which is combined with a convolutional neural network (CNN) to sort traffic signs. CNN may be utilized to perform a variety of computer vision tasks due to its high recognition rate. TensorFlow is used to implement CNN and Keras. In this, we are able to identify the symbol with more accuracy rate and with lesser time.

4.Methodology

Each of the 43 folders in our dataset folder represents a different class. The folder's size ranges from 0 to 42. We iterate over all of the classes using the OS module, appending images and their labels to the data and labels list. Finally, we organized all of the images and labels into lists (data and labels). To feed the model, we must turn the list into NumPy arrays. The data has the shape (39209, 30, 30, 3), indicating that there are 39,209 images of 3030 pixels and that the last three indicate that the data comprises colored images (RGB value).

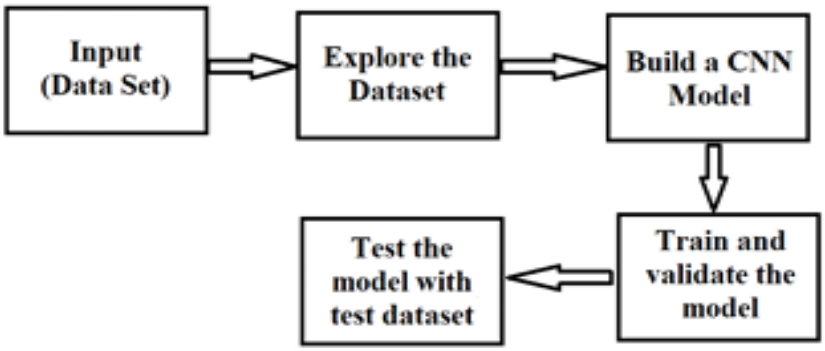


Figure 1 – Dataset Block Diagram

Importing Libraries

- Pandas – Use to load the data frame in a 2D array format.
- NumPy – NumPy arrays are very fast and can perform large computations in a very short time.
- Matplotlib – This library is used to draw visualizations.
- OpenCV – This library mainly focused on image processing and handling.
- TensorFlow – It provides a range of functions to achieve complex functionalities with single lines of code.

We'll use a CNN model to classify the photos into their appropriate groups (Convolutional Neural Network). For picture categorization, CNN is the best option. After constructing the model architecture, we use model.fit to train the model (). We tried batch sizes of 32 and 64. With 64 batches, our model fared better. The accuracy was stable after 10 epochs. On the training dataset, our model had a 98.47% accuracy rate. We visualize the graph for

accuracy and loss using matplotlib. The details connected to the image path and their appropriate class labels are contained in a Label.csv file in our dataset. Using pandas, we extract the image path and labels. Then, in order to forecast the model, we must scale our photographs to 3030 pixels and create a NumPy array with all of the image data. In this model, we were able to attain a 98.47% accuracy rate.

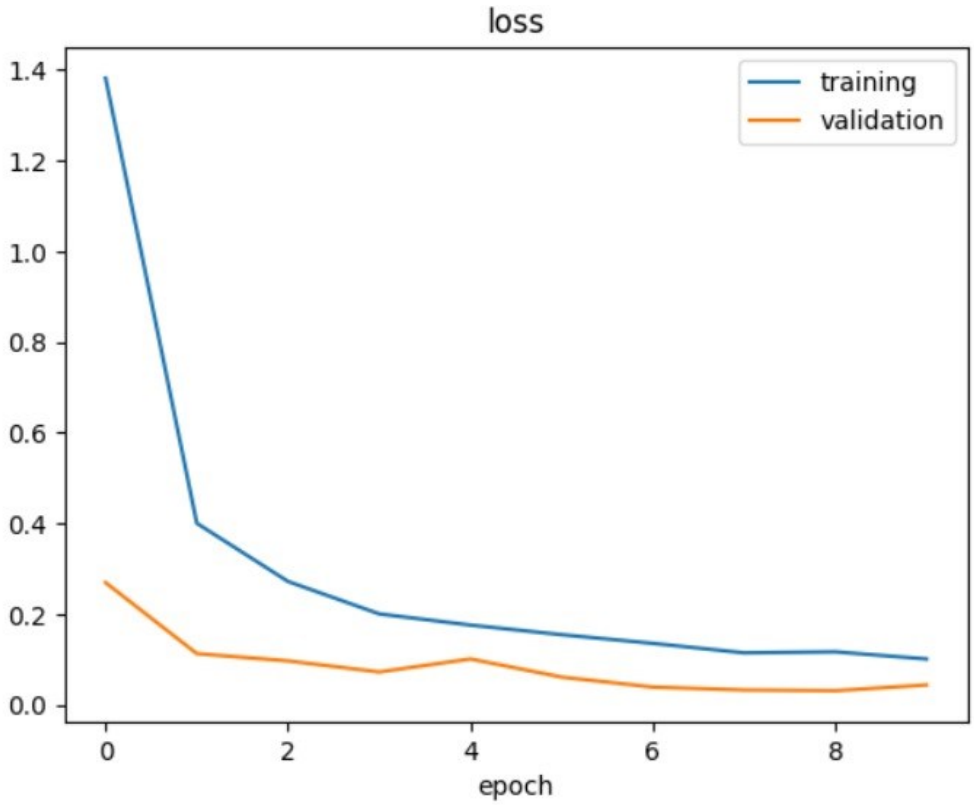


Figure 2

The graph shows the loss, not accuracy, of a training and validation process over a number of epochs . The x-axis is labeled epoch, and the y-axis is labeled loss. Epoch refers to one complete pass through the training data. In machine learning, loss refers to how well a model performs on a specific task. Lower loss indicates better performance. The goal of training a model is to reduce the loss. The graph shows that the training loss (blue line) decreases as the number of epochs increases. This suggests that the model is learning the training data well and performing better on the training task. The validation loss (red line) also decreases initially, but then starts to plateau after a few epochs. This suggests

that the model may be starting to overfit the training data. Overfitting is a modeling problem where a machine learning model performs well on the training data, but poorly on unseen data . When a model overfits, it has learned the specific patterns of the training data too well and does not generalize well to new data. In conclusion, the graph shows that the model is learning the training data well, but it may be starting to overfit the data as indicated by the flattening validation loss curve.

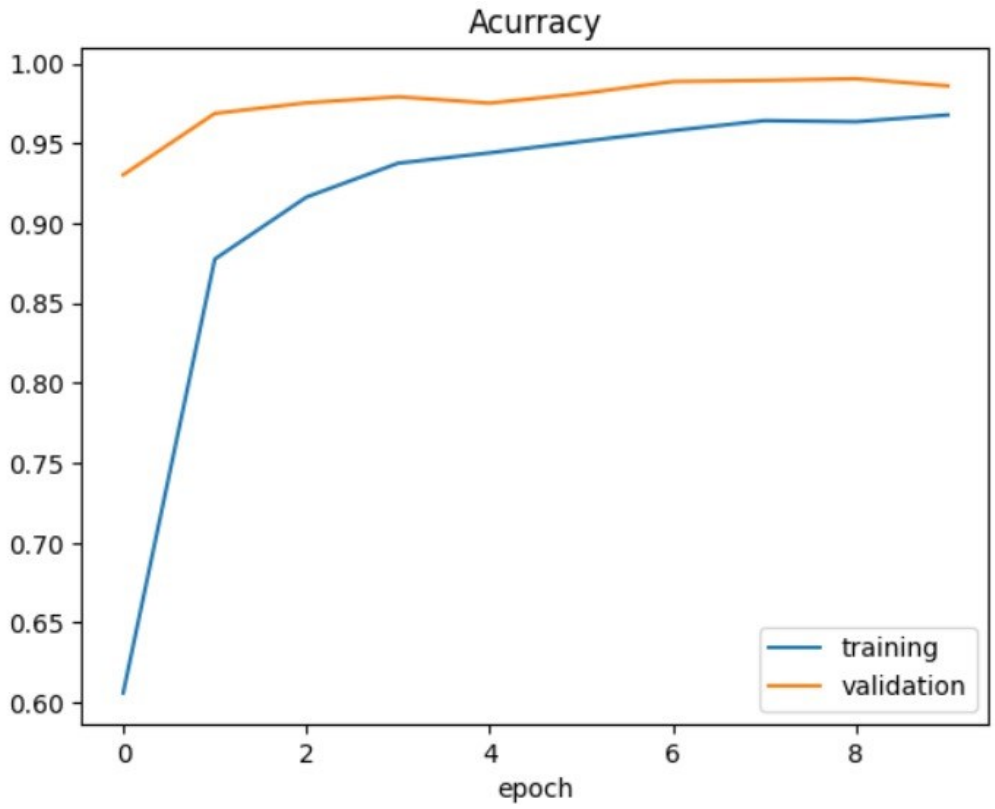


Figure 3

This graph plots the accuracy of a training and validation process . The x-axis is labeled epoch, and the y-axis is labeled accuracy. Epoch refers to one complete pass through the training data. The graph shows that the training accuracy increases as the number of epochs increases. This suggests that the model is learning the training data well. However, the validation accuracy increases slower than the training accuracy, and starts to level off after a few epochs. This suggests that the model may be starting to overfit the training data. Overfitting is a modeling problem where a machine learning model performs well on the training data, but poorly on unseen data.

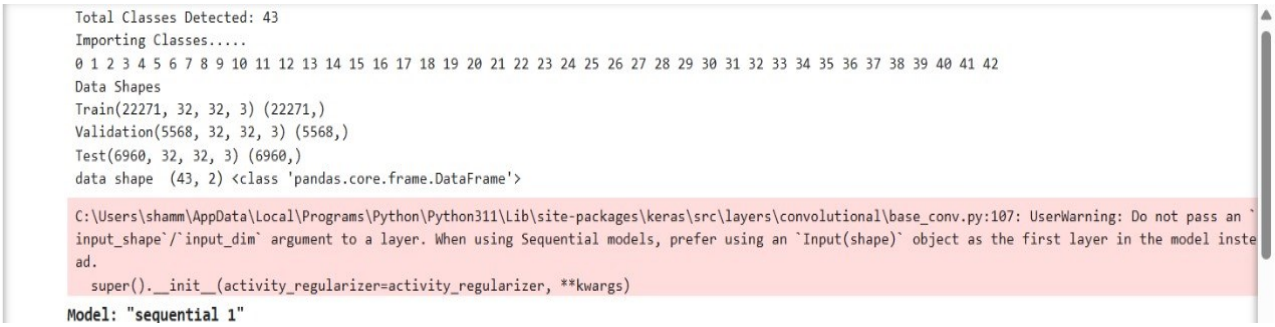


Figure 4

All the necessary details which are essential for the python to run the program are imported. The images which are already available in the dataset are retrieved with their tables. The list which are numerical values of the image is being converted into NumPy arrays. Dataset is split into training and testing. The labels are converted into one hot encoding. Building of the model is done with the use of the data set which is already available and we got in the dataset splitting process. The complication of the models is done. For checking the accuracy graphs are plotted with the help of the result we got from the solution. The accuracy of the result with the help of the trained dataset which we have already split into testing and training.

5.Result and Discussion

With the help of the given algorithms and others python libraries, we have successfully achieved the result of identifying the traffic signs in different environments. So, with the help of the given dataset from Kaggle, we achieved training sets based on the dataset. With the use of the model.h5 classifier file is created. With that classifier file, we are able to get the results quickly. The system is capable of classifying the input images with a 98.47% accuracy rate.

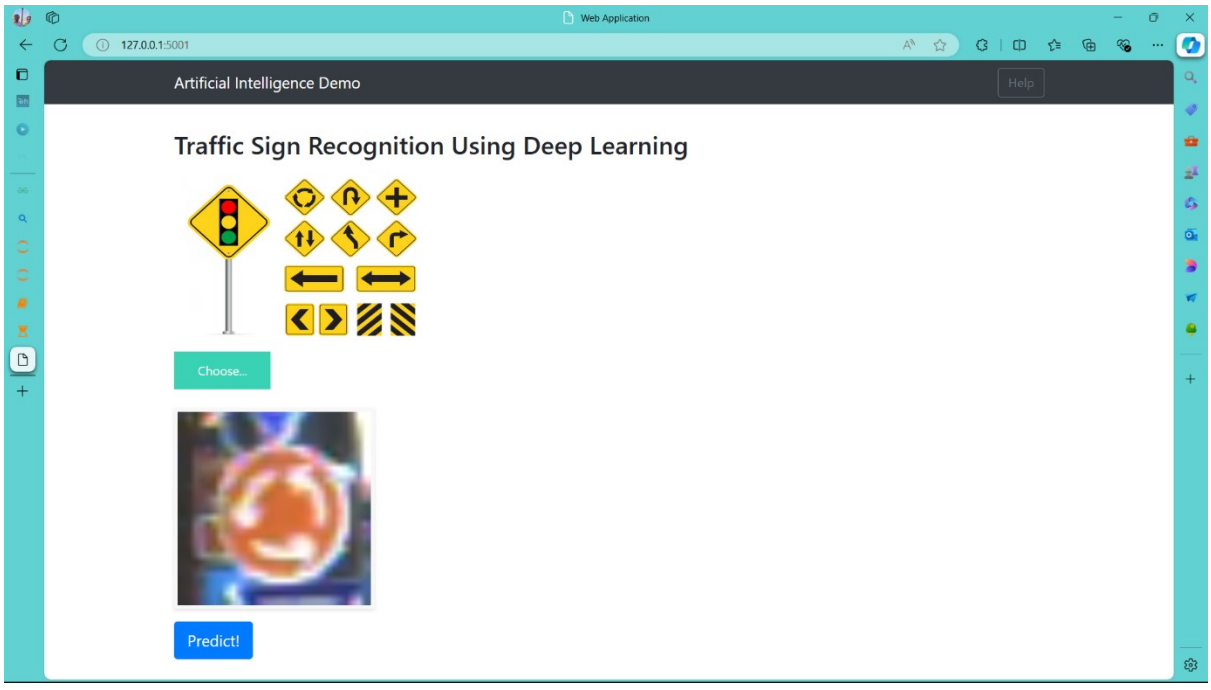


Figure 4

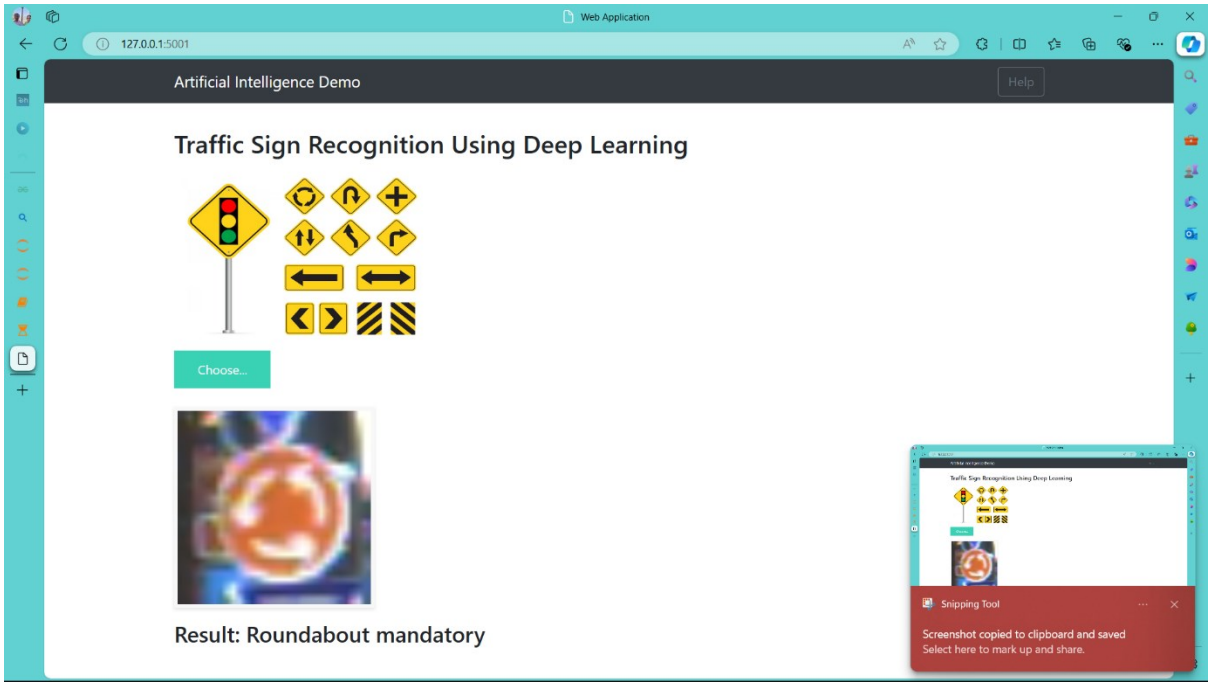


Figure 5

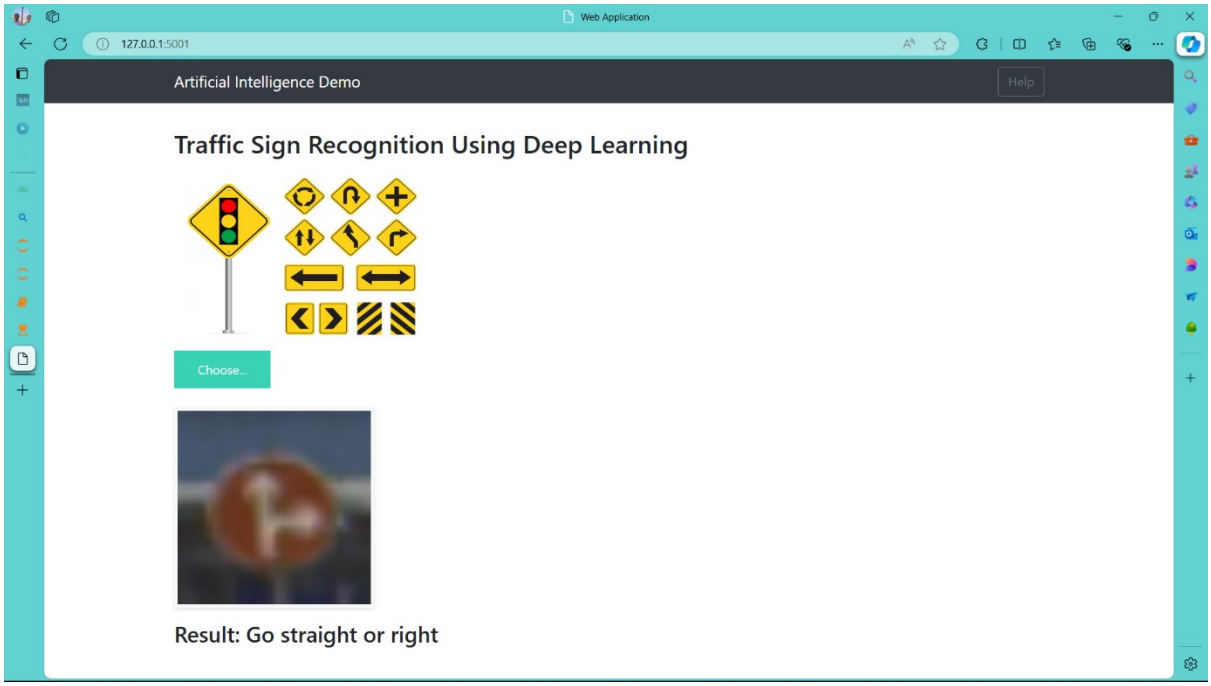


Figure 6

That image is given as input to the system. With help of a convolutional neural network (CNN) and Keras, images are classified. Dataset is used for training our algorithm, there are different sets of data present in the dataset. There are data of different climatic conditions and different tampered sign boards. During those difficult situations our algorithm with help of CNN and Keras, we are able to achieve a high accuracy rate for classifying the traffic signs.

6.Conclusion

In this paper, a traffic sign recognition method on account of deep learning with help of a convolutional neural network (CNN) is proposed, which mainly aims at different traffic signs. By using image pre-processing, the dataset from Kaggle, traffic sign detection, recognition, and classification, this method can effectively detect and identify traffic signs. With help of these results, we can identify the traffic sign. It helps the user in two ways, while the user is in manual mode it displays the result on the dashboard screen and while the car driver is set to automatic it helps the car to drive safely by identifying the traffic signs. The test result displays that the accuracy of this method is very high.

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